

Iteration skipping via Padé approximation

for deep zoom rendering of the Mandelbrot set

We want to speed up a pixel's orbit computation by *skipping* blocks of iterations. The standard method (**BLA**, *bivariate linear approximation*) replaces a block by an *affine* map; we show here how a *rational* Padé [1/1] map captures one extra order, at the cost of a benign pole.

1 Setting: perturbation

Let $C \in \mathbb{C}$ be the centre of the view. The **reference orbit** is computed once, in high precision:

$$Z_0 = 0, \quad Z_{n+1} = Z_n^2 + C.$$

Each pixel is a small offset $c = \delta c$ around C . Its orbit is $Z_n + z_n$, where the **delta** z_n follows the perturbed iteration

$$z_{n+1} = 2Z_n z_n + z_n^2 + c, \quad z_0 = 0. \quad (1)$$

The point: z_n stays tiny, hence computable in low precision even when $Z_n + z_n$ requires hundreds of digits. The *linear* term $2Z_n z_n$ dominates; the *quadratic* term z_n^2 is what the approximations will treat differently.

2 BLA: the affine approximation

The BLA idea is to drop z_n^2 over a block, which linearises (1). One step becomes $z_{n+1} \approx A z_n + B c$ with the **seed**

$$A = 2Z_n, \quad B = 1.$$

Two consecutive steps (first x , then y) compose trivially, since composing two affine maps stays affine:

$$A_z = A_y A_x, \quad B_z = A_y B_x + B_y. \quad (2)$$

This is what enables a hierarchical table of blocks of sizes 1, 2, 4, ...

Validity domain. Dropping z_n^2 is legitimate as long as it stays small relative to the linear term:

$$|z_n^2| \ll |2Z_n z_n| \iff |z_n| \ll |2Z_n| = |A|.$$

With a machine threshold ε , the block is valid on the disc

$$\boxed{|z_n| < R_{\text{aff}} = \varepsilon |A|} \quad (3)$$

3 Padé [1/1]: the rational approximation

3.1 Seed: reproducing the quadratic term

Rather than discarding z_n^2 , we look for a rational fraction that *reproduces* it. We posit the *bivariate* form (linear in c , rational [1/1] in z):

$$z_{n+1} \approx \frac{A z_n + B c}{1 + D z_n}. \quad (4)$$

Its series expansion in z_n (with c fixed) is

$$\frac{Az}{1 + Dz} = Az - ADz^2 + AD^2z^3 - \dots$$

Matching term by term with the *exact* step $2Z_n z + z^2 (+c)$:

$$\underbrace{A = 2Z_n}_{\text{order } z^1}, \quad \underbrace{-AD = 1 \Rightarrow D = -\frac{1}{2Z_n}}_{\text{order } z^2}, \quad \underbrace{B = 1}_{\text{term in } c}.$$

Padé seed. $A = 2Z_n, \quad B = 1, \quad D = -\frac{1}{2Z_n} = -\frac{1}{A}.$

The rational form reproduces z^2 *exactly* (the affine one drops it). Its leading error is the parasitic cubic term $+AD^2z^3 = z^3/(2Z_n)$. It introduces a **pole** at $z = -1/D = 2Z_n = A$: precisely the boundary where linearisation was already losing all meaning. The pole thus *encodes* the local divergence.

3.2 Composition: the Möbius structure

At $c = 0$, map (4) is a Möbius transform, representable by a 2×2 matrix:

$$T(z) = \frac{Az}{1 + Dz} \longleftrightarrow M = \begin{pmatrix} A & 0 \\ D & 1 \end{pmatrix}.$$

Composing two blocks amounts to *multiplying* their matrices:

$$M_y M_x = \begin{pmatrix} A_y & 0 \\ D_y & 1 \end{pmatrix} \begin{pmatrix} A_x & 0 \\ D_x & 1 \end{pmatrix} = \begin{pmatrix} A_y A_x & 0 \\ D_x + A_x D_y & 1 \end{pmatrix}.$$

The composition rule reads off directly. The coefficient B (c -dependence) propagates to first order as in the affine case:

Composition of two Padé blocks (x before y):

$$A_z = A_y A_x, \quad B_z = A_y B_x + B_y, \quad \boxed{D_z = D_x + A_x D_y}.$$

The first two rules are identical to the BLA (2). The novelty is D_z : a sum weighted by the flow derivative A_x , encoding the block's *curvature* where the affine map retains only the *slope*.

Remark. The exact composition produces cross terms in cz and c : the $[1/1]$ form is not strictly closed. But in deep zoom $|c|$ is infinitesimal, and these higher-order-in- c terms truncate harmlessly; the form then closes exactly on the rules above.

3.3 Validity domain: the gain

Compare the relative error per step.

$$\text{Affine: } \frac{|z^2|}{|Az|} = \frac{|z|}{|2Z|} \quad \Bigg| \quad \text{Padé: } \frac{|AD^2z^3|}{|Az|} = \left(\frac{|z|}{|2Z|} \right)^2.$$

Requiring the error to stay below ε yields the two validity radii:

$$R_{\text{aff}} = \varepsilon |A| \quad \longrightarrow \quad \boxed{R_{\text{Padé}} = \sqrt{\varepsilon} |A|}. \quad (5)$$

Reach gain. $\frac{R_{\text{Padé}}}{R_{\text{aff}}} = \frac{1}{\sqrt{\varepsilon}} \approx 10^8$ in double precision ($\varepsilon \sim 10^{-16}$), $\approx 4 \times 10^3$ in single precision.

In practice, a Padé block stays valid for much larger $|z|$ — notably *near escape*, where z grows and the affine map immediately falls back to step-by-step. This is the only quantitative argument in favour of the method.

3.4 Application, with a safeguard

At run time, apply the accumulated block (A, B, D) to the current delta:

$$\text{den} = 1 + Dz, \quad z \leftarrow \begin{cases} \frac{Az + Bc}{\text{den}}, & |\text{den}| \geq \theta, \\ 2Zz + z^2 + c \text{ (exact step)}, & |\text{den}| < \theta, \end{cases}$$

the threshold $\theta \sim 10^{-2}$ preventing blow-up near the pole.

4 Indispensable safeguards

Three conditions, discovered during implementation, govern the *correctness* of the result — independently of the affine/Padé choice. Ignoring them produces massive artefacts, identical in both modes and insensitive to ε .

1. **Never skip starting from $Z \approx 0$.** The reference starts at $Z_0 = 0$, so $A_0 = 2Z_0 = 0$ annihilates any accumulated block. Since *rebasing* periodically returns to Z_0 , force an exact step while $|Z| < \delta$.
2. **Minimum skip $L_{\min} = 4$.** Over 2 steps from the origin, the delta equals *exactly* c^2 : a purely quadratic result that *no* linear map can represent. Let exact perturbation handle short intervals.
3. **Reference more precise than the delta.** Storing Z_n in single precision (~ 7 digits) drowns the delta signal as soon as you zoom. A *double-single* encoding ($Z = Z_{\text{hi}} + Z_{\text{lo}}$) is the minimum.

Honesty note. The perturbation framework, the affine BLA, and the safeguards are established and documented (Heiland-Allen, *mathr.co.uk*; Zhuoran, *fractalforums*). The Padé part — seed $D = -1/2Z$, composition $D_z = D_x + A_x D_y$, radius $\sqrt{\varepsilon}|A|$ — is an original derivation, *not validated* by the fractal-rendering community, and to be checked against experiment before any production use.